

10	C	The minute hand takes 1 hour to go round the clock once. $\omega_m = \frac{2\pi}{60 \times 60} \text{ rad s}^{-1}$ The hour hand takes 12 hours to go round the clock once. $\omega_h = \frac{2\pi}{12 \times 60 \times 60} \text{ rad s}^{-1}$ $\frac{v_m}{v_h} = \frac{r_m \omega_m}{r_h \omega_h} = \left(\frac{1.5r_h}{r_h} \right) \left(\frac{12 \times 60 \times 60}{60 \times 60} \right) = 18$
11	C	Option C (correct):